

Mohamed Hesham

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SUMMARY

Software Engineer with 6+ years of Python backend experience across enterprise automation, distributed systems, and LLM-powered applications. Shipped event-driven automation frameworks, RAG pipelines, and observability backends serving enterprise clients and 500+ end users. MS in Computer Science and AWS Certified Solutions Architect knowledge

SKILLS

Languages: Python, SQL, C++(Familiar)

Frameworks & Tools: FastAPI, Flask, RESTful APIs, PostgreSQL, Docker, Linux, Git, CI/CD

Distributed Systems & Data: Distributed Pipelines, Parallel Scaling, High-Performance Computing (HPC), Dask

Applied AI: Intelligent Agents, Semantic Search, Vector Indexing, Retrieval-Augmented Generation (RAG)

Cloud & Certifications: AWS (EC2, S3, Lambda, RDS), CCNA

EXPERIENCE

Community Dreams Foundation, Remote

Apr 2026 - Present

Backend Developer

- Develop and maintain Python REST APIs and backend services for a nonprofit platform built on FastAPI and PostgreSQL.
- Containerize and ship backend services with Docker through automated CI/CD workflows in a production environment.
- Drive engineering quality through system design reviews, automated testing, and service monitoring.

Texas State University, San Marcos, TX

Jan 2024 - Dec 2025

Graduate Assistant

- Designed, developed, and deployed a full-stack Python/Flask production web application serving 70+ active users, owning real-time data flows and feature delivery across the full application lifecycle.
- Automated end-to-end data processing pipelines that replaced manual reporting workflows, improving data consistency and reliability for university decision-makers.

J&M Jewelry, San Marcos, TX

Jan 2025 - Apr 2025

Software Engineer Intern

- Engineered an LLM-powered extraction pipeline that converted unstructured product data into validated structured records, raising processing accuracy by 25% for 500+ monthly users.
- Exposed the pipeline through a FastAPI backend with RESTful endpoints and automated its release cycle with CI/CD testing and deployment workflows.

Orange Business, Cairo, Egypt

Apr 2021 - Dec 2023

VoIP Incident Management

- Diagnosed complex SIP call setup and teardown failures by tracing signaling flows and response codes (4xx/5xx/6xx), validating SDP offer/answer exchanges, and isolating registration or NAT traversal faults on SBC and IP-PBX platforms.
- Developed custom Python scripts to automate recurring diagnostics, including automated SIP log parsing and alarm triage, reducing manual root-cause analysis time by 30%.
- Resolved persistent one-way audio, jitter, and call drop incidents by analyzing RTP captures via Wireshark; corrected DSCP markings, firewall policies, and NAT misconfigurations across MPLS WAN links to restore mean opinion scores (MOS).

PROJECTS

LangChain Multi-Tool AI Agent | Python, LangChain, OpenAI, Tool Calling, FastAPI

[GitHub](#)

- Built a multi-tool AI agent that routed natural-language requests to specialized tools using structured input schemas and a modular tool registry.
- Designed persistent session context and execution logging to improve reliability, traceability, and behavior across multi-turn chats.

FastAPI RAG Pipeline with LLM | Python, FastAPI, LLaMA 3.1, LangChain, ChromaDB, Ollama

[GitHub](#)

- Built an end-to-end retrieval and inference pipeline using FastAPI and ChromaDB with optimized chunking and vector embedding strategies, improving retrieval precision across document collections.
- Deployed LLaMA 3.1 8B locally via Ollama with 4-bit quantization, managing resource-constrained inference infrastructure with full observability over query latency and retrieval accuracy.

Distributed XGBoost Training & Performance Profiling | Python, Dask, XGBoost, SLURM, OpenMP

[GitHub](#)

- Engineered distributed ML training pipelines on TACC Lonestar6 HPC across multi-node GPU environments using Dask and SLURM.
- Profiled parallel scaling across worker configurations and identified XGBoost internal threading as the primary bottleneck.

EDUCATION

Texas State University, San Marcos, TX

Dec 2025

Master of Science, Computer Science | Unity Scholarship Winner | GPA: 3.92

Arab Academy for Science and Technology, Cairo, Egypt

Jun 2019

Bachelor of Engineering, Electronics and Communications